



Week 2

Project Scope Management, Planning and Scheduling

**UWE
Bristol**

University
of the
West of
England

Content

- Scope Management
- Work Breakdown Structure (WBS)
- Consolidated Planning
- Network Diagram
- Change management



Scope management



Defines the project boundaries

- Clarifies what work is included and what is excluded
- Sets clear start-to-finish limits for the team

Aligns on objectives

- States the outcomes the project must achieve
- Links deliverables to goals and stakeholder value

Creates a shared understanding

- Documents scope so everyone works from the same baseline

Enables successful delivery

- Provides the basis for plans, estimates, and milestones
- Supports decision-making during change

Challenges & solutions



Unclear requirements

- Turn goals into acceptance criteria; confirm with examples.
- Baseline scope early; log assumptions and open questions.



Stakeholder expectations

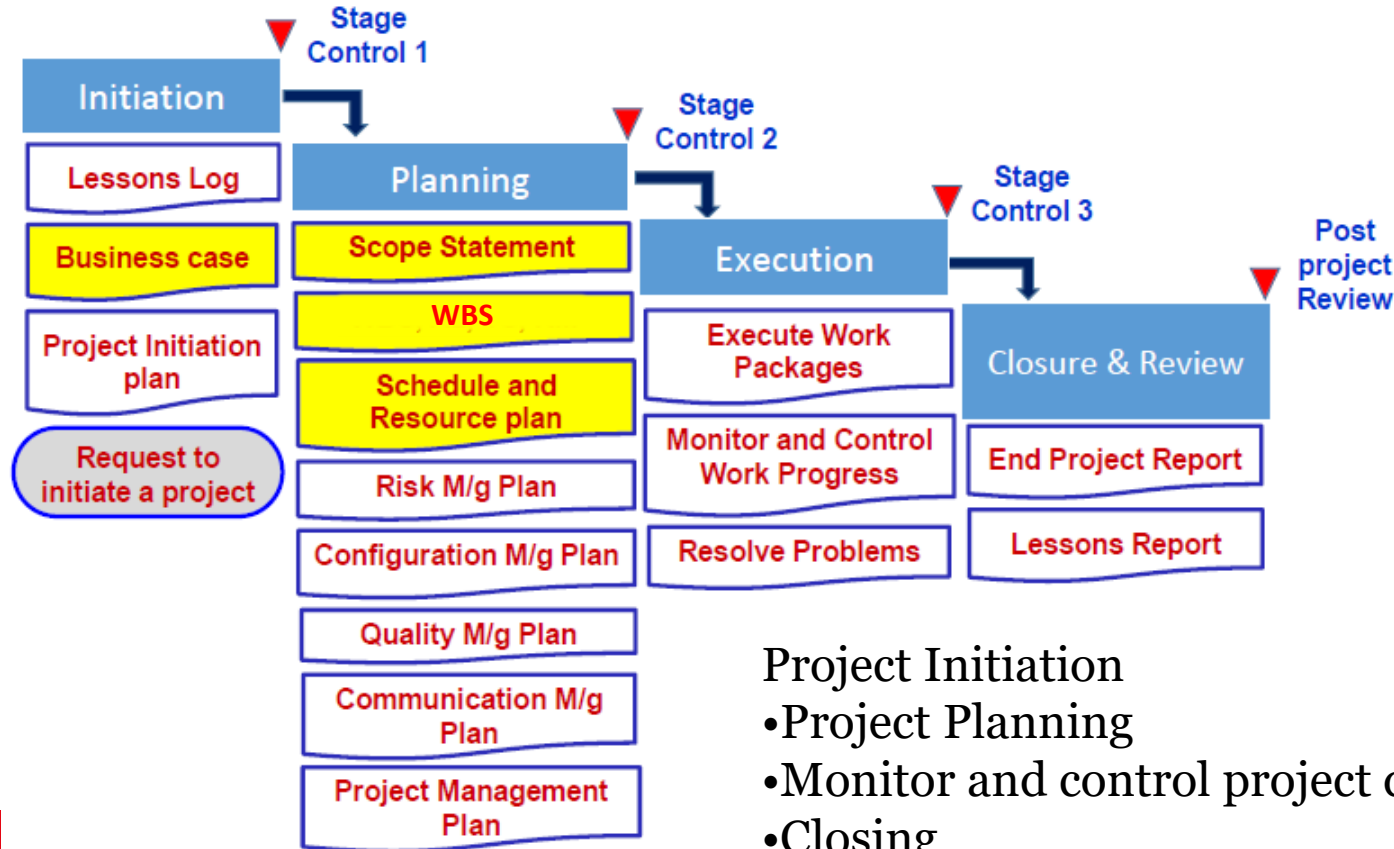
- Align on priorities, trade-offs, and success metrics upfront.
- Use a simple change process: impact, approve, update plan.



Tools + communication

- Single source of truth: backlog, scope statement, decisions.

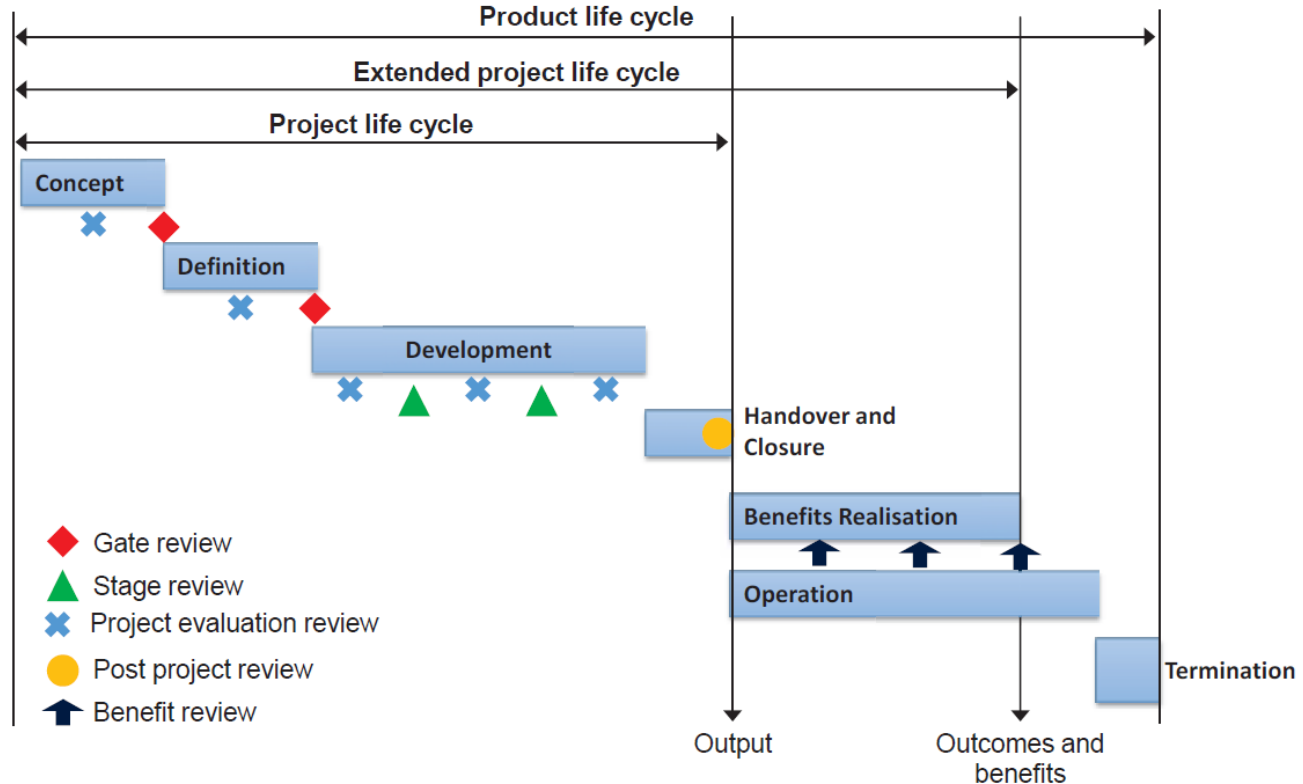
Project Life-cycle



Project Initiation

- Project Planning
- Monitor and control project deliverables
- Closing

Project Life-cycle: The APM Methodology



Project Life-cycle: The APM Methodology

Concept Phase

The phase develops an initial idea and creates an outline business case or project brief and schedule.

Key actions include:

- Customer consultation
- Needs and requirements identification and definition
- Stakeholder analysis
- Identification of benefits
- Production of approved need statement

Definition Phase

The preferred solution is identified and ways of achieving it are refined and detailed plan(s) in the form of the Project Management Plan (PMP)

Key actions include:

- Refined estimates of timescales and costs
- Risks identification and assessment
- Identification of required resource
- Project team formation
- The creation of the PMP and detailed business case

Project Life-cycle: The APM Methodology

Benefits Realisation

project may include a benefits realisation phase. This will depend to a large extent whether the project is delivered internally to the organisation or to an external client

Key actions include:

- Implement the benefits realisation plan
- Conduct benefits realisation reviews
- Life extension and value adding initiatives
- Lessons learned initiatives

Why Scope

Inadequate **understanding and agreement of scope** will lead to frustration near the end of the project & poor project performance.

Typical examples

Poor estimates based on not understanding the totality of the work.

Lack of scope change management because scope was not properly defined to begin with.

- Issues occurring because of **poor risk management**.
- Missing the schedule due to **poor discipline**.
- Not understanding all the **stakeholders** involved.



Scope Management

What does the term **Scope** mean within a project context?

“The processes required to ensure that the project includes:

All the work required,

and

Only the work required, to complete the project successfully”

The **Scope** must answer the question –

Within the agreed
specification and
requirements

“What is the **project's** end product”.



Why to manage Scope

Schedule and budget can't be managed if scope is out of control
(Triple Constraint)

The Scope is used to manage all stakeholders expectations



Typical Scope Management

1. **Requirements Management:** Gathers and assesses **stakeholder wants and needs**. Requirements describe stakeholder's wants and needs but do not determine exactly how they will be met,
2. **Requirement Development:** Take the **stakeholder's requirements** and investigates how they will be achieved,
3. **Change Management:** Deals with the **transformation** of the business-as-usual that is necessary to utilise outputs and realise benefits,
4. **Configuration Management: Monitors and documents the development of products.** Making sure that approved changes are recorded and superseded versions archived.
5. **Programme:** Typically within a project plan is in place
6. **Agreement: Defines and agrees on the means to demonstrate compliance** with each requirement of the project or product. Also describes the organisation to deliver the project and the detailed planning.

Scope Misalignment

So what happens when scope is not controlled?

1. What consequences should we expect?
2. What is the Cost and Time?



Projects get derailed very easily.

Repeat Requirement Error

Florida

1. New Fire Truck too big for the Boylston St. Station
2. Resolution add new front to station
3. Knock on problem was that there was not enough clearance to access the street



Others

1. Boston – Engine didn't fit historic station
2. Tusculumbia Alabama – Engine too tall
3. Dunfermline, Scotland – Engine too big for streets
4. Saranac Lake NY – Engine too tall & too heavy
5. Elkhville, Ill – New station not built yet
6. London, Ontario, Canada – Engine too tall (measurement error)
7. Edmondson, AR – Engine too wide
8. Morant Bay Jamaica – Engine too wide for streets

Grenfell Tower, Requirements, Budget, Competence

Grenfell Tower is a 24-storey residential tower block in North Kensington in London, England.

- It was completed in 1974, as part of the first phase of the Lancaster West Estate.
- Grenfell Tower underwent a major renovation in 2015-16 including external cladding
- The fire caused 72 deaths
- Burned for about 60 hours until finally being extinguished. More than 250 London Fire Brigade firefighters and 70 fire engines
- Cost-driven design changes affecting critical outcomes.

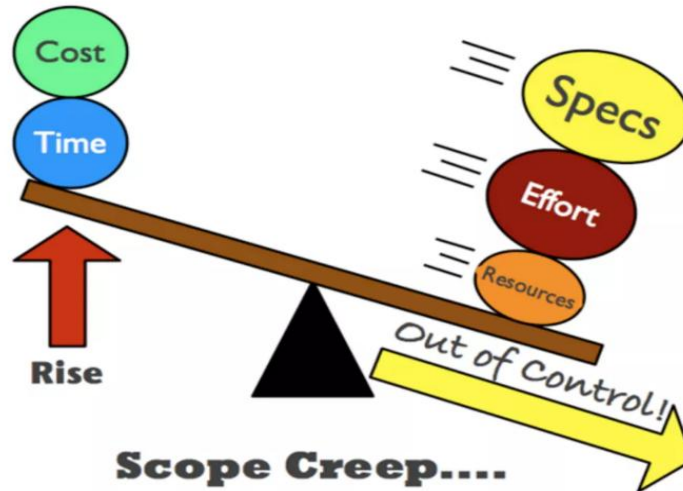
https://www.youtube.com/watch?v=-h4Eu-Ynysk&ab_channel=SkyNews



Scope Creep A Priority

An ever present threat in industry is Scope Creep, driven by:

- The customer is always right (changes happen mid programme)
- key project stakeholders changing requirements, or sometimes from internal miscommunication and disagreements



How to manage Scope

Changes to scope can be **controlled** - Controlling those changes in scope via a change management process. This involves:

Comparing actual work performance measurements with the baseline -
“How different is the current project from the original plan?”

Deciding on each change requests, i.e. whether corrective or preventive action is needed

If approved change requests affect the project's overall scope and cost baseline, then the scope statement, Work Breakdown Structure (WBS), and/or cost baseline is updated and sent out to stakeholders.

Key Actions to control Scope

Initial statement signed by all parties:

- Agreed Time Cost Quality goals of project.
- Agreed applicable Requirements
- Agreed change Control Mechanism including cost allocation
- Agreed resolution process

During Project

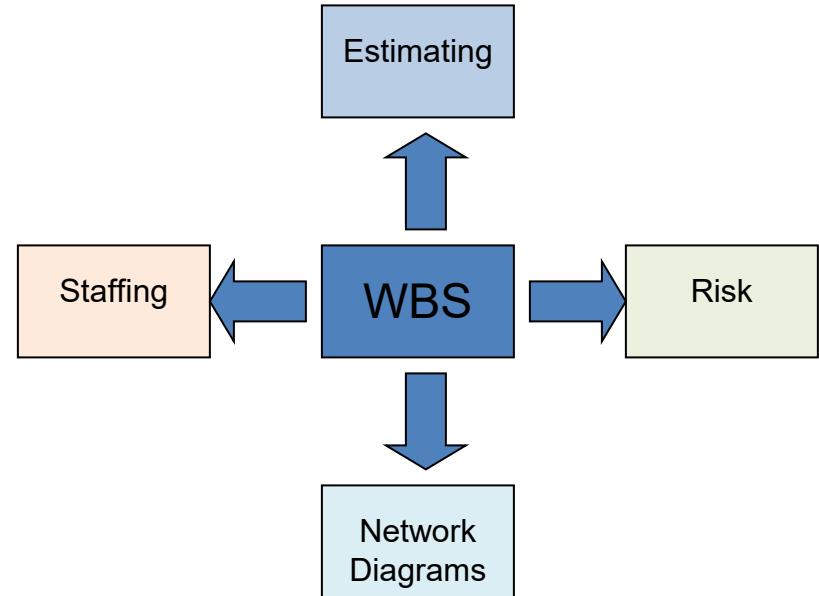
- Design Reviews held and agreed by all parties
- Live Change control process – each change signed off by all parties
- Final Design Review giving technical approval to product

Project Closure

Signed agreement by all parties that product has been delivered as required by Initial Statement plus applicable Changes.

Work Breakdown Structure (WBS)

- Foundation of the project, all planning and controlling is based on the WBS
- Identifies all work to be performed, if it is not in the WBS it does not need to be done
- Graphical picture of work



Definitions

You define the work to be done
for each activities

The WBS (Work Breakdown Structure)

- Organisational Chart or Technique, the HOW, WHERE is a deliverable-oriented breakdown of a project into smaller components

CBS (Cost Breakdown Structure), which is a hierarchical map of the costs in your project. The cost breakdown structure works off a WBS but it maps project deliverables and tasks, by determining the costs of those deliverables and tasks.

This tends to be the starting point of
most project, define the organisation

The OBS (Organisational Breakdown Structure)

- **Who is responsible for WHAT** defined in WORKPACKAGES (WPs)
- Managers are defined at each WBS node. One and only one manager per node.

You assign your resources to the
work to be done

The RBS (Resource Breakdown Structure)

- **WHO does WHAT.** Resources are defined at the task level. One or more resources per task

Consolidated Plan

Consolidated Planning: the process of documenting and integrating the key components of a project into a controlling document, also known as a Project Management Plan (PMP). The purpose of a consolidated plan is to serve as the primary reference for managing a change initiative.

Scope, Schedule, Resource requirements, Budgets, Risks, opportunities and issues, and Quality requirements



Contents of the PMP

The Project Management Plan (PMP) is the master document of any project, defining the “why”, “what”, “how”, “who”, “when” and “where” of the project

Introduction/Background: Reasons behind the project; Project objectives from business case.

Scope: Detail of how the project scope be managed from definition to close-down

Scheduling: The Work Breakdown Structure, Cost Breakdown Structure; information regarding project milestones, Gantt charts and the level of any networks used.

Resource Management: Organisation Breakdown Structure, Resource Management Plan

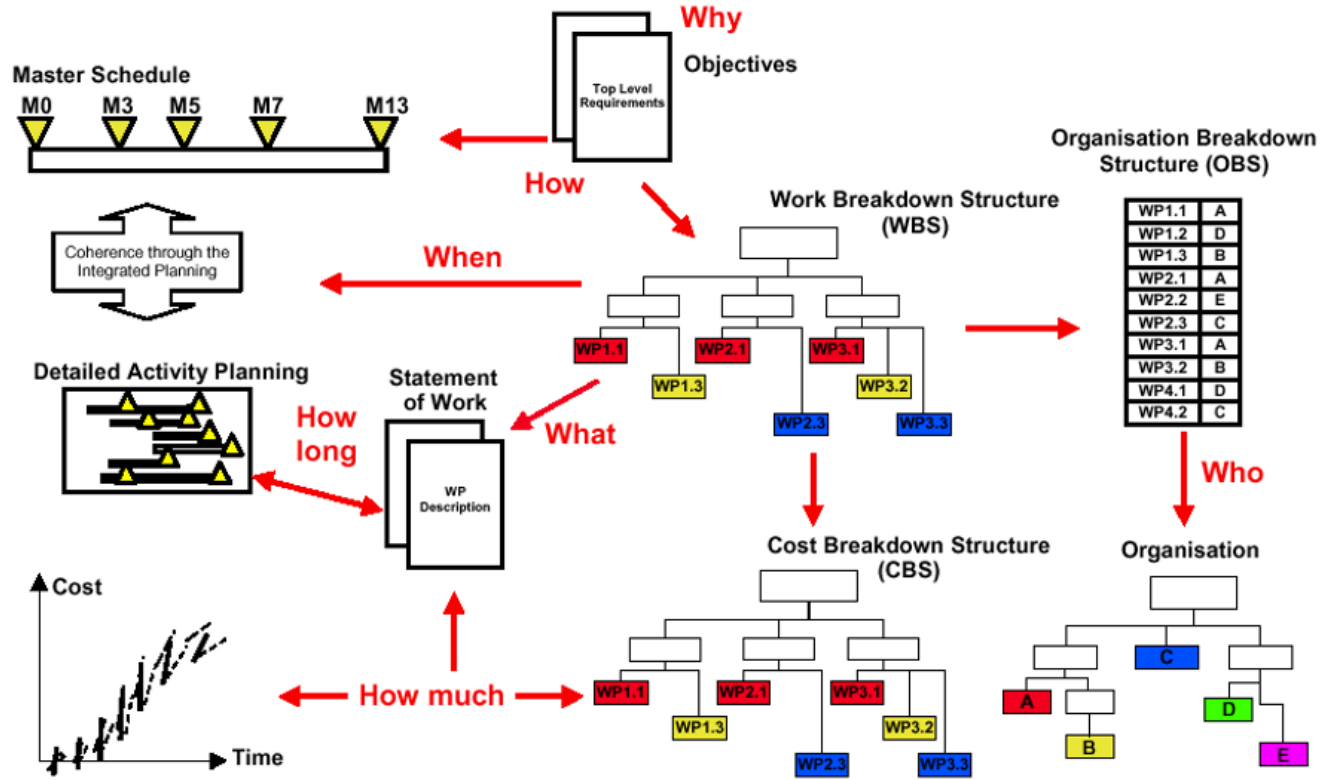
Requirements Management Plan: Plan of all requirements required by customer

Performance Management Plan: How the project will measure, monitor and control performance against the requirements timescales, costs and quality criteria.

Risk, Change and Configuration Management Plans: Detailed information on how the project will record and manage and deal with these issues

Other plans - Quality Management & Safety Management Plan: Both important plans containing key activities needed to be carried out

Relationship between PMP



The different components of the overall Project Management System

Stages of Planning

Breakdown the project into packages of work in accordance with the Work Breakdown Structure.

Identify the project activities and estimate their duration.

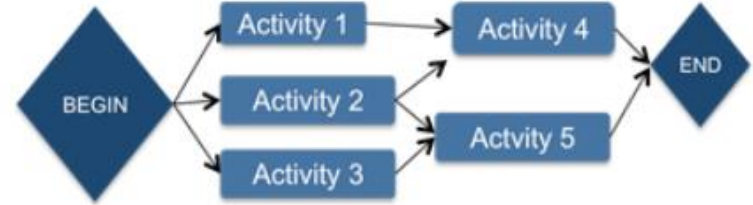
Estimate the likely cost of each activity, and derive the Cost Breakdown Structure (CBS).

Draw up a schedule (i.e. time phased activities) and identify the critical project activities. (Critical Path)

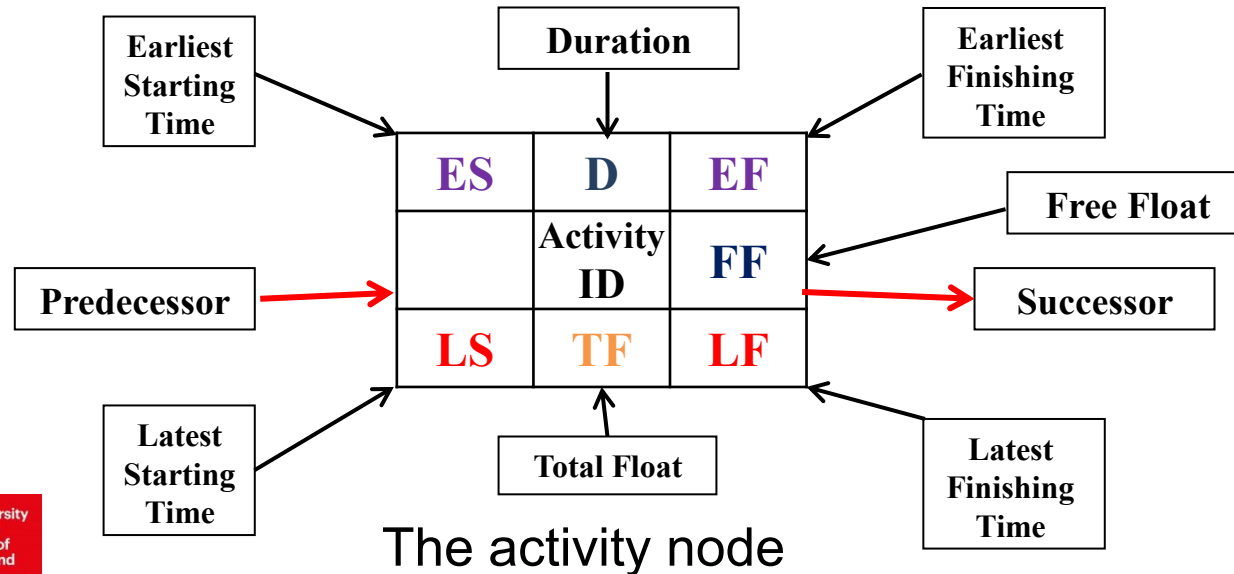
Activity on Node (AON) Network

Draw an **Activity on Node (AON)** network to represent the activities of your project

- i. Activities are represented by **nodes**
- ii. The dependencies between activities are indicated by **arrows**

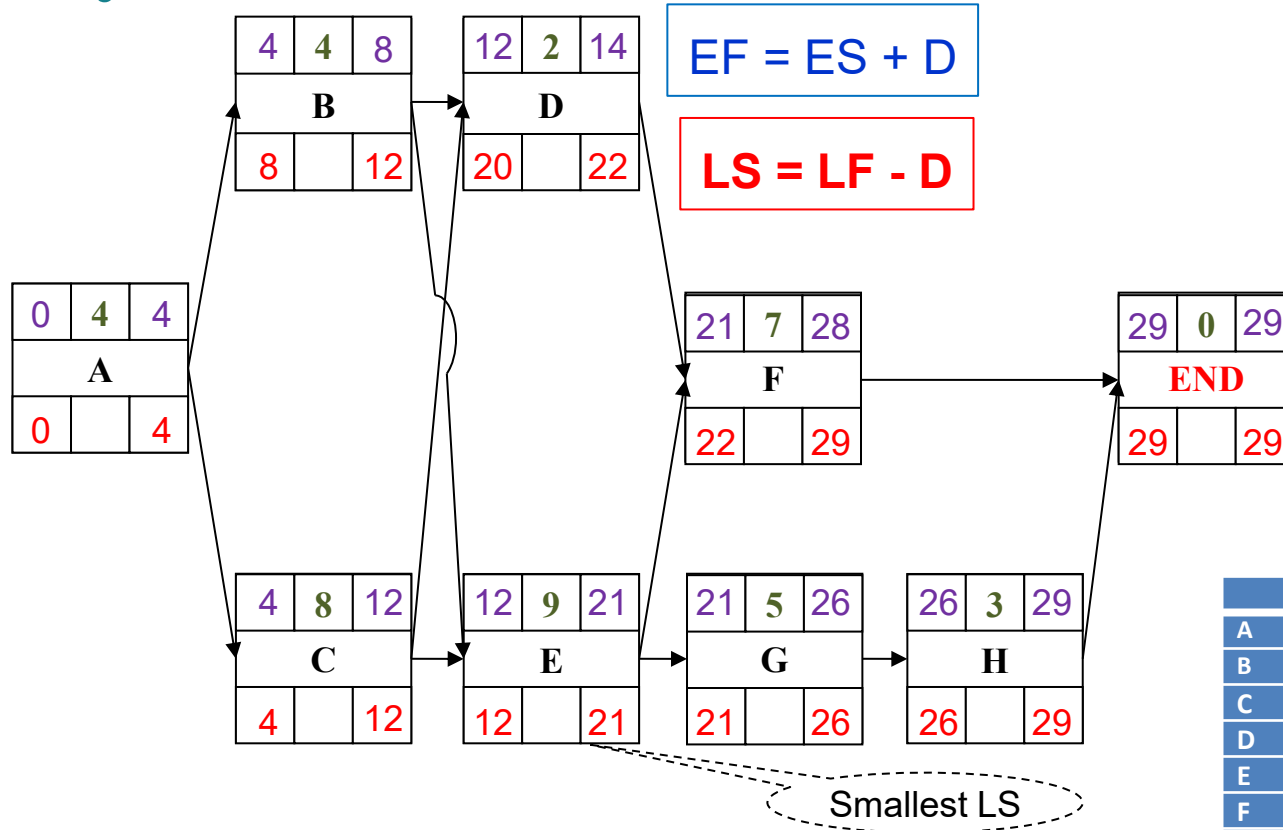


[Link: How to draw a PDM network diagram](#)



Activity on Node (AON) Network

ES	D	EF
Activity ID	FF	
LS	TF	LF



	Days	Predecessor
A	4	-
B	4	A
C	8	A
D	2	B,C
E	9	B,C
F	7	D,E
G	5	E
H	3	G

Example of calculating the Late Activity Times

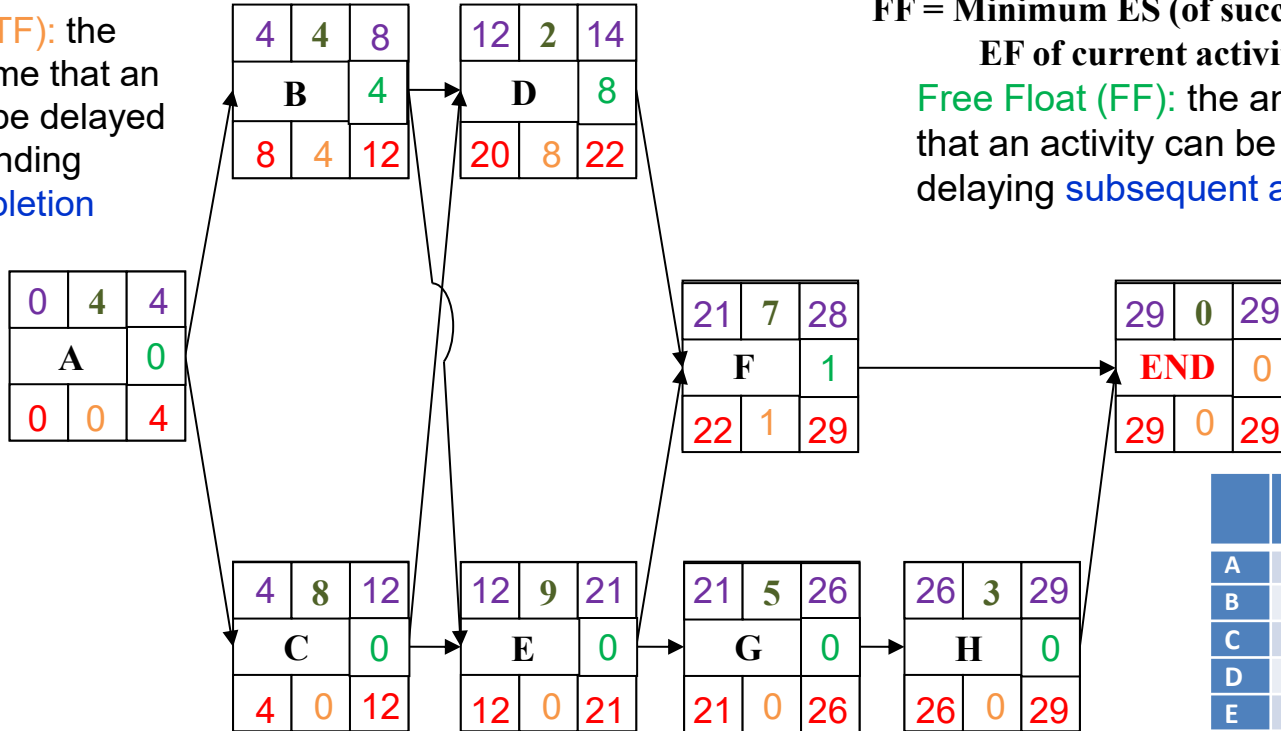
Activity on Node (AON) Network

$$TF = LS - ES = LF - EF$$

Total Float (TF): the amount of time that an activity can be delayed without extending project completion time

FF = Minimum ES (of successors) – EF of current activity

Free Float (FF): the amount of time that an activity can be deferred without delaying subsequent activities

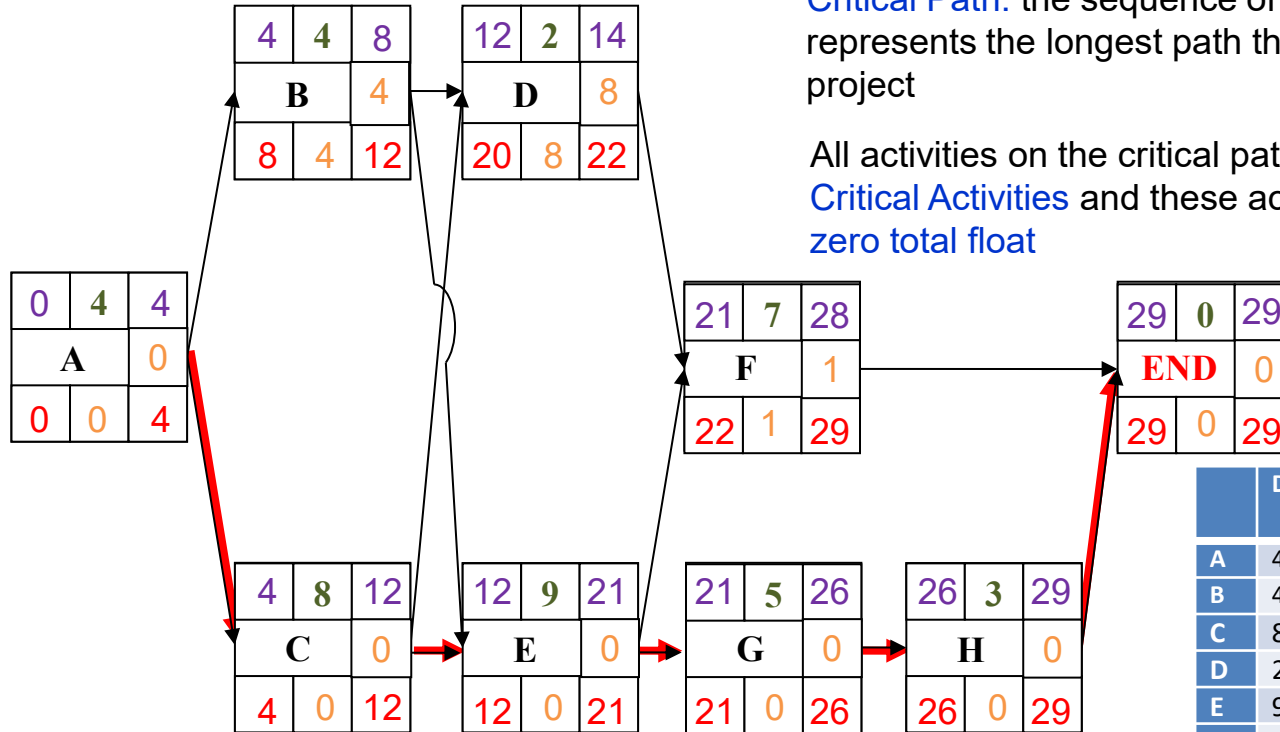


Example of calculating the Free Float of an activity

	Days	Predecessor
A	4	-
B	4	A
C	8	A
D	2	B,C
E	9	B,C
F	7	D,E
G	5	E
H	3	G

Activity on Node (AON) Network

ES	D	EF
Activity ID		FF
LS	TF	LF



Critical Path: the sequence of activities that represents the longest path through a project

All activities on the critical path are called **Critical Activities** and these activities have zero total float

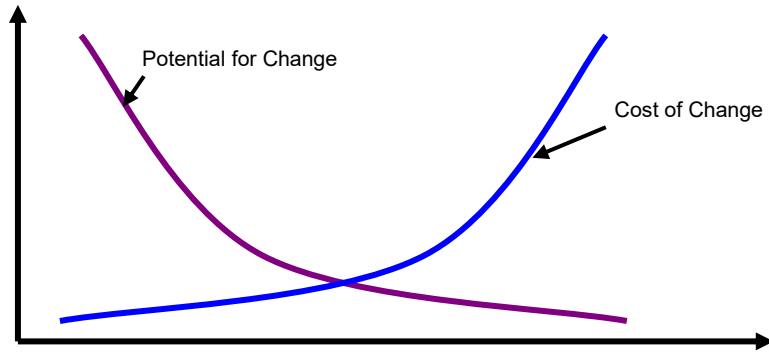
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F	7	D,E
G	5	E
H	3	G

A network with the Critical Path highlighted

Change Management

Change Management controls any changes to time, cost and quality/performance

Changes can occur at any time, however the impact of a change late on in a project is usually more costly than if the change is made early in the project's life



Evaluate change

- Identify owner
- Describe change
 - ✓ Impact on schedule (Time)
 - ✓ Impact on resource (Cost)
 - ✓ Impact on requirement (Quality)
- Budget estimate
- Risk analysis

Project success takeaways



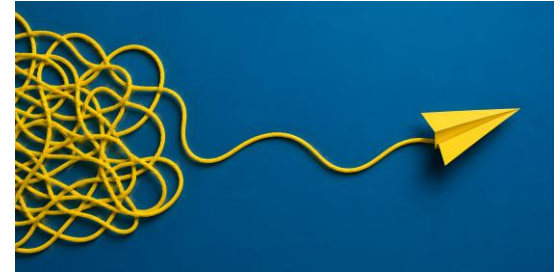
Keep scope clear and bounded

- Define deliverables, exclusions, and success criteria up front.
- Review scope at milestones to prevent **scope creep**.



Communicate early, document always

- Capture decisions, assumptions, and changes in one shared place.
- Use brief weekly check-ins to surface risks fast.



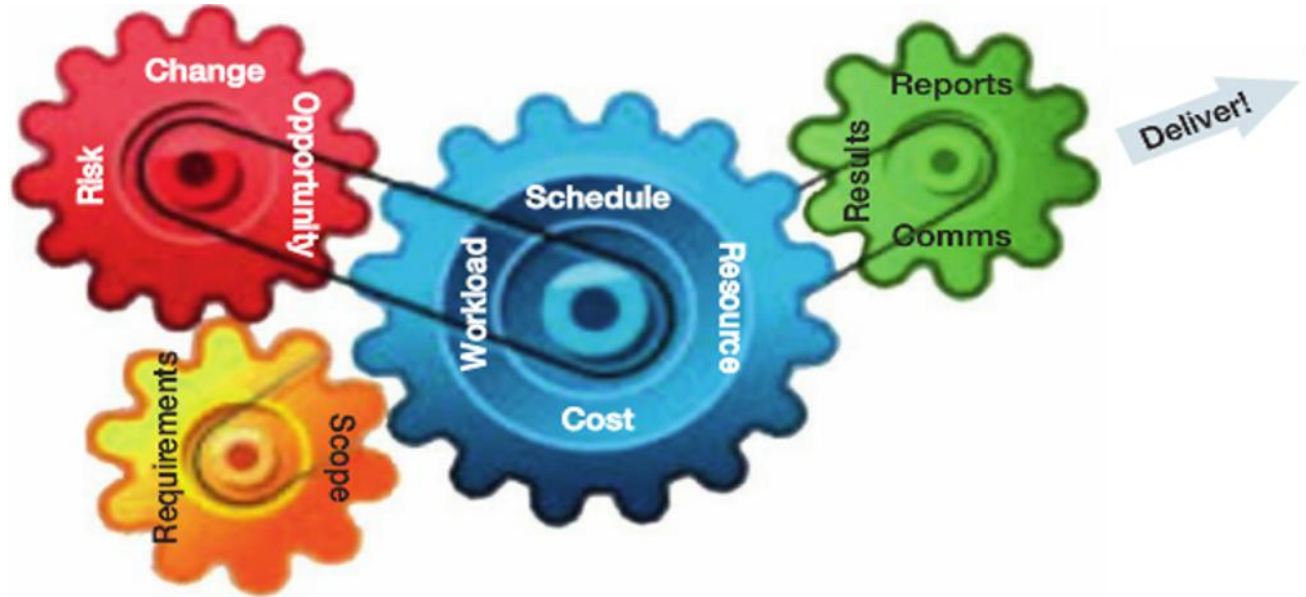
Empower the team to manage change

- Set a simple change process: request, impact, approve, update.

Summary

Starting a project is the beginning of a journey.

- Initiation/Idea
- Triple Constraint
- Scoping
- Planning and Scheduling
 - WBS, Cost Breakdown Structure, Work Packages, Critical Path
- Change Management
 - Control and evaluation, impact of the change



*Thank
you*

